



**MERIDIAN JUNIOR COLLEGE**  
JC2 Preliminary Examinations 2017  
Higher 2

CANDIDATE  
NAME

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CIVICS  
GROUP

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INDEX  
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## H2 BIOLOGY

**9744/01**

Paper 1 Multiple Choice

**21 September 2017**

**1 hour**

Additional Materials: Multiple Choice Answer Sheet

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### READ THESE INSTRUCTIONS FIRST

**Do not open this booklet until you are told to do so.**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Write your name, civics group and index number on the Multiple Choice Answer Sheet provided.

There are thirty questions in this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice in soft pencil on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

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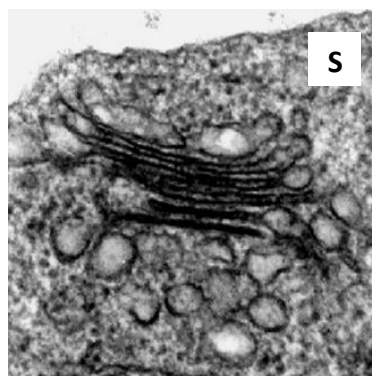
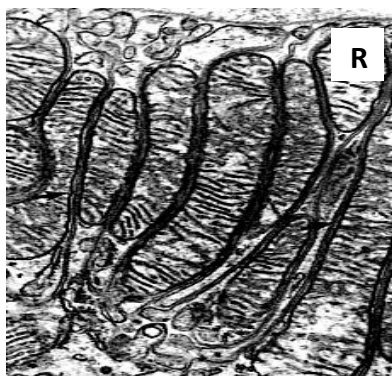
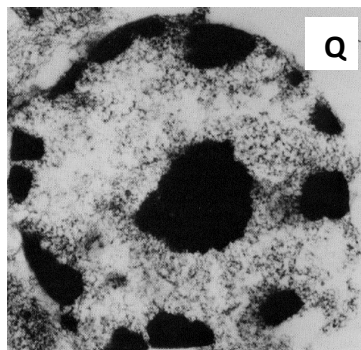
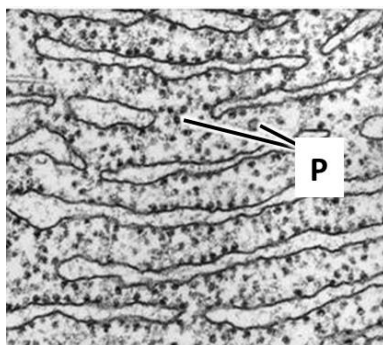
This paper consists of **19** printed pages.

**[Turn over]**

### QUESTION 1

Cell fractionation is a method used to study cell components. It is achieved by taking a number of cells and breaking their cells surface membranes to release the contents of the cells into a buffer solution, and then subjecting the contents to gentle homogenization to preserve the integrity of the organelles.

In zonal centrifugation, the suspension of cell contents is placed on top of a sucrose density gradient. The tube is then placed in a centrifuge and spun at high speed.

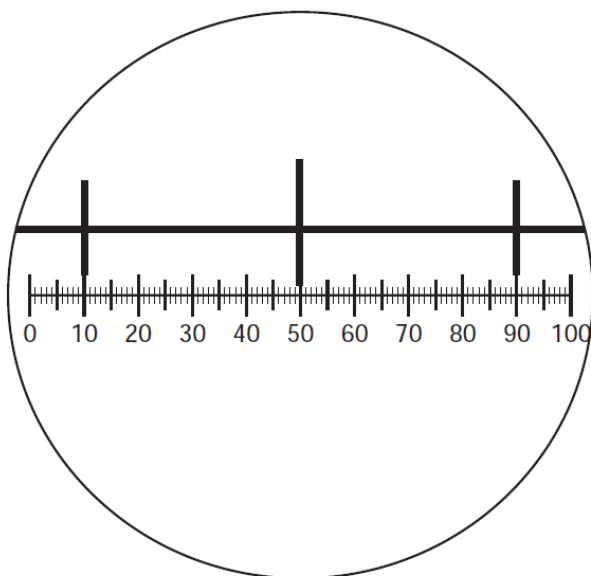


Which of the following options shows the positions of the organelles after centrifugation from the top to the bottom of the sucrose density gradient?

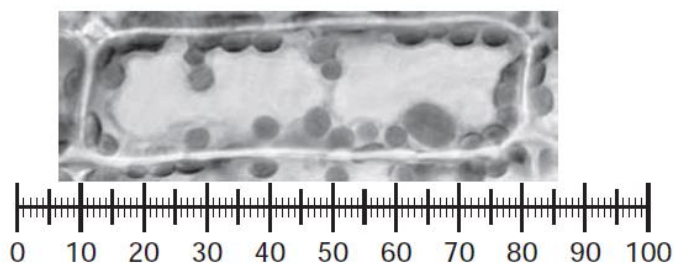
|          | top <span style="float: right;">→ bottom</span> |          |          |          |
|----------|-------------------------------------------------|----------|----------|----------|
| <b>A</b> | S                                               | R        | Q        | P        |
| <b>B</b> | R                                               | S        | P        | Q        |
| <b>C</b> | <b>P</b>                                        | <b>R</b> | <b>S</b> | <b>Q</b> |
| <b>D</b> | Q                                               | S        | R        | P        |

## QUESTION 2

The diagram shows a stage micrometer on which the small divisions are 0.1 mm. It is viewed through an eyepiece containing a graticule.



The stage micrometer is replaced by a slide of a plant cell.



What is the diameter of a chloroplast?

- A.** 0.5 mm      **B.** 10  $\mu\text{m}$       **C.** 50  $\mu\text{m}$       **D.** 100  $\mu\text{m}$

### QUESTION 3

An antibiotic inhibits the formation of cross-links between the molecules that form cell walls in bacteria.

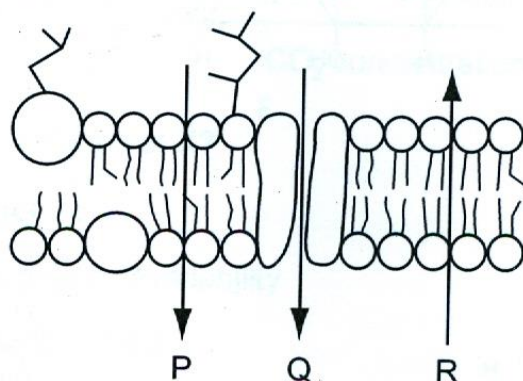
Which statements explain why bacteria are killed by the antibiotic?

1. The bacterial cell is destroyed by osmotic lysis.
2. The cellulose molecules cannot form hydrogen bonds.
3. The cell wall is no longer selectively permeable.

A 1 and 2 only      B 2 and 3 only      **C 1 only**      D 2 only

### QUESTION 4

The diagram shows the cell surface membrane of an actively respiring cell in a tissue that has been placed in a solution of glucose with a lower water potential than that of the tissue cells.



What correctly describe the movements of molecules across the cell surface membrane shown by arrows P, Q and R?

|          | P                          | Q                                       | R                         |
|----------|----------------------------|-----------------------------------------|---------------------------|
| A        | diffusion of glucose       | diffusion of oxygen                     | diffusion of water        |
| B        | diffusion of oxygen        | diffusion of water                      | diffusion of glucose      |
| C        | diffusion of water         | active transport of glucose             | diffusion of oxygen       |
| <b>D</b> | <b>diffusion of oxygen</b> | <b>facilitated diffusion of glucose</b> | <b>diffusion of water</b> |

### QUESTION 5

Which biological molecules always contain the element nitrogen?

- A. glycine, cellulose, mRNA  
B. collagen, DNA, lipids  
**C. enzymes, mRNA, HIV genome**  
D. membrane proteins, starch, tRNA

### QUESTION 6

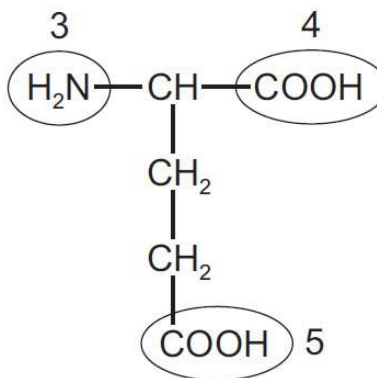
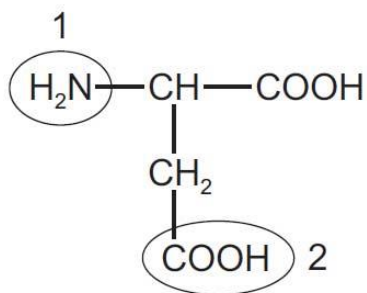
Which features allow a cellulose molecule to be adapted for its function?

1. Long chains of  $\beta$ -glucose molecules have multiple branches.
2. Many hydrogen bonds are formed between adjacent chains.
3. It is insoluble in water.
4. There is a high proportion of the amino acid glycine, which has a very small side chain.

A. 2 and 3 only      B. 3 and 4 only      C. 1, 2 and 3 only      D. 2, 3 and 4 only

### QUESTION 7

The diagrams show the structures of two amino acids, each of which has two carboxylic acid groups ( $-\text{COOH}$ ).



Which groups form the bonds that maintain the configuration of  $\alpha$ -helices?

A. 1 and 4      B. 1 and 5      C. 2 and 3      D. 2 and 5

### QUESTION 8

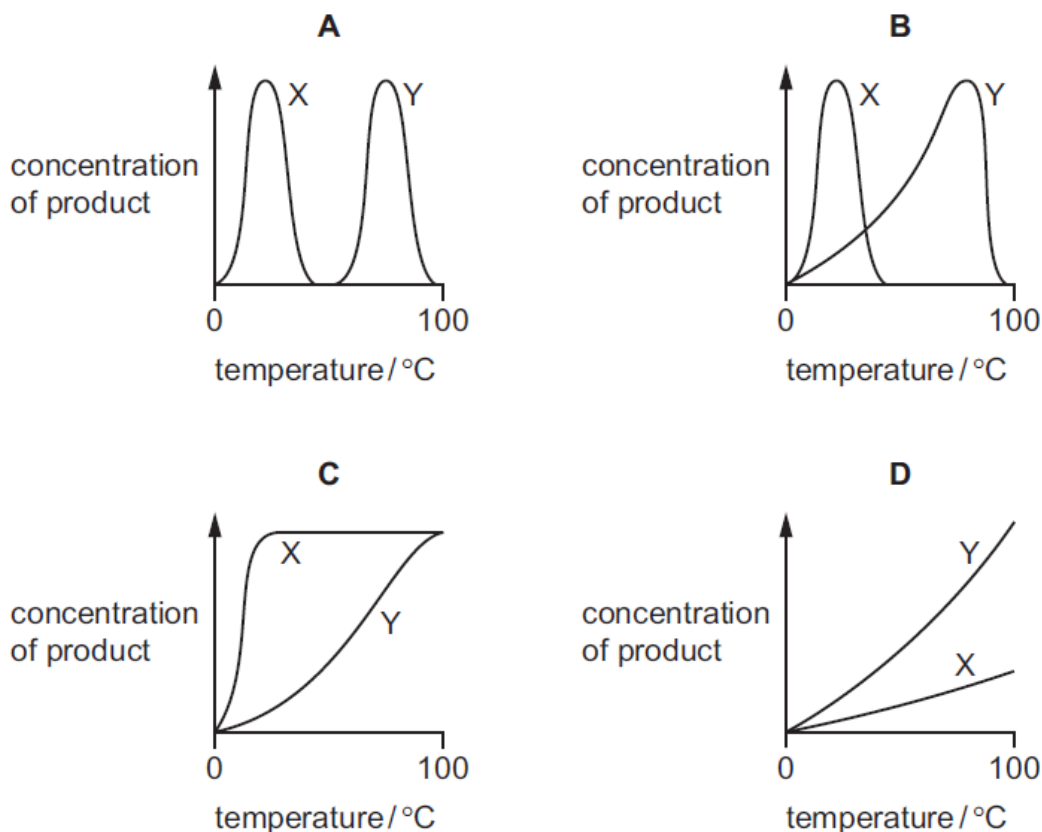
Two enzymes, **X** and **Y**, were used in an experiment.

Enzyme **X** was from bacteria that live in rivers and lakes at temperatures from 5°C to 20°C.

Enzyme **Y** was from bacteria that live in hot water springs at temperatures from 40°C to 85°C.

The experiment measured the concentration of product produced by each enzyme at temperatures between 0°C and 100°C after 5 minutes.

Which graph shows the results? **(B)**



### QUESTION 9

Which statements about the cell cycle are correct?

1. Heterochromatin takes a longer time than euchromatin to replicate during S phase.
2. Different cells have different durations of the cell cycle because the length of G<sub>1</sub> phase is the most variable.
3. DNA is repaired in each checkpoint to ensure the integrity of DNA molecules.

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

### QUESTION 10

The ends of a eukaryotic chromosome contains a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

Then cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

In some cells, telomerases are present as a counter-measure.

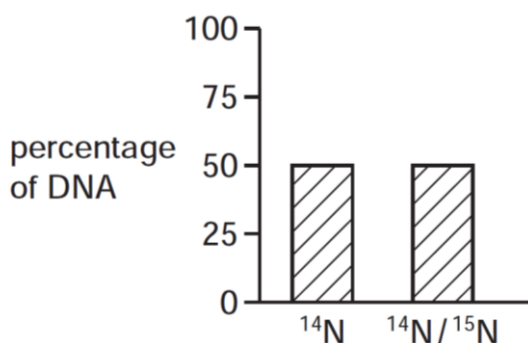
Which description of the consequence of the loss of telomeres and of the role of telomerase reverse transcriptase is correct?

|           | <b>Consequence of the loss of telomeres</b>                           | <b>Role of telomerase reverse transcriptase</b>                               |
|-----------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>A.</b> | The cells will synthesise different proteins.                         | Uses RNA as a template to make single-stranded DNA.                           |
| <b>B.</b> | Mitosis will be halted at the G2 checkpoint.                          | Inhibits the loss of telomeres from DNA during semi-conservative replication. |
| <b>C.</b> | The number of mitotic divisions the cell can undergo will be limited. | Uses RNA as a template to make single-stranded DNA.                           |
| <b>D.</b> | Lead to the end-to-end fusion of chromosomes together.                | Inhibits the loss of telomeres from DNA during semi-conservative replication. |

### QUESTION 11

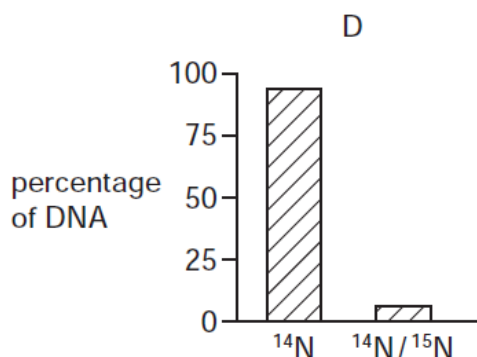
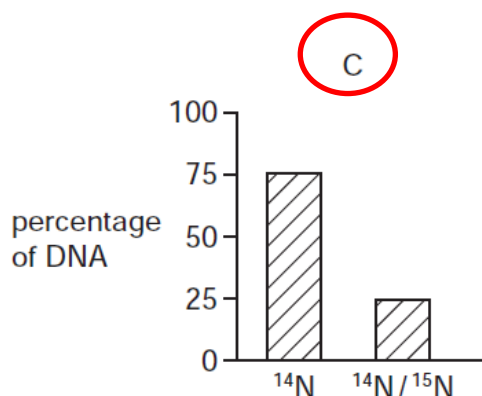
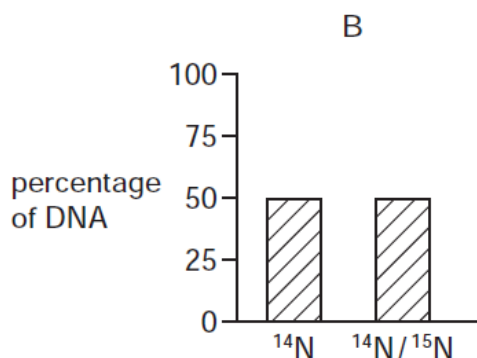
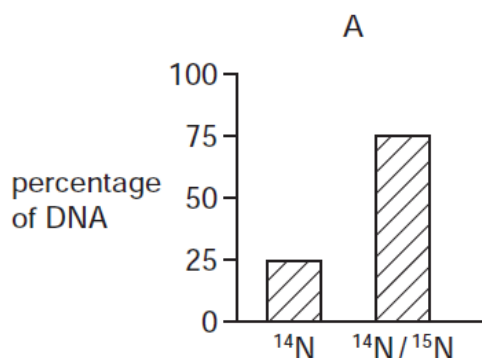
Bacteria were grown in a medium containing  $^{15}\text{N}$ . After several generations, all of the DNA contained  $^{15}\text{N}$ . Some of these bacteria were transferred to a medium containing the common isotope of nitrogen,  $^{14}\text{N}$ . The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amount of  $^{14}\text{N}$  and  $^{15}\text{N}$ .

Some bacteria from the medium with  $^{15}\text{N}$  were transferred into a medium of  $^{14}\text{N}$ . The bacteria were allowed to divide twice. The graph shows the percentage of  $^{14}\text{N}$  and  $^{15}\text{N}$  in the DNA of these bacteria.



Some bacteria from the medium with  $^{15}\text{N}$  were transferred into a medium of  $^{14}\text{N}$ . The bacteria were allowed to divide three times.

What would be the percentage of  $^{14}\text{N}$  and  $^{15}\text{N}$  in the DNA extracted from these bacteria?





**QUESTION 12**

Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are:

lys-glu-thr-ala-ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

| first position | second position          |                          |                            |                           | third position   |
|----------------|--------------------------|--------------------------|----------------------------|---------------------------|------------------|
|                | U                        | C                        | A                          | G                         |                  |
| U              | phe<br>phe<br>leu<br>leu | ser<br>ser<br>ser<br>ser | tyr<br>tyr<br>STOP<br>STOP | cys<br>cys<br>STOP<br>trp | U<br>C<br>A<br>G |
| C              | leu<br>leu<br>leu<br>leu | pro<br>pro<br>pro<br>pro | his<br>his<br>gln<br>gln   | arg<br>arg<br>arg<br>arg  | U<br>C<br>A<br>G |
| A              | ile<br>ile<br>ile<br>met | thr<br>thr<br>thr<br>thr | asn<br>asn<br>lys<br>lys   | ser<br>ser<br>arg<br>arg  | U<br>C<br>A<br>G |
| G              | val<br>val<br>val<br>val | ala<br>ala<br>ala<br>ala | asp<br>asp<br>glu<br>glu   | gly<br>gly<br>gly<br>gly  | U<br>C<br>A<br>G |

Which event occurs to explain the information given above?

- 1 The first amino acid on the polypeptide chain is removed in post-translational modification.
- 2 The first codon is removed from the mRNA transcript in post-transcriptional modification.
- 3 The mRNA binds to the rRNA in the second codon position during translation.
- 4 There is no tRNA with an anticodon complementary to the first codon.

**A. 1 only**

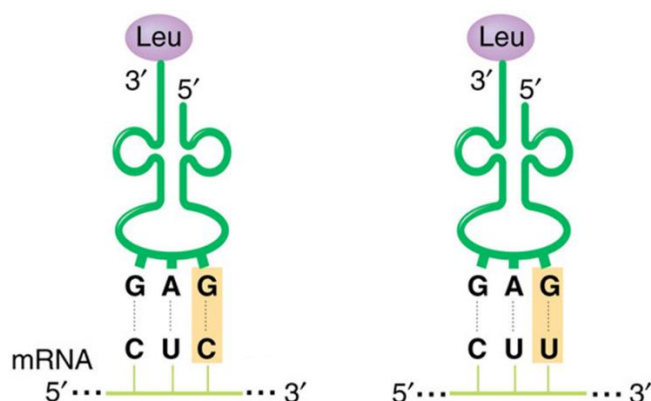
**B. 3 only**

**C. 1 and 2**

**D. 1, 2 and 4**

### QUESTION 13

Figure below shows two aminoacyl-tRNA and a corresponding complementary region on the mRNA based on the Wobble hypothesis.



Which of the following are possible conclusions that can be made from the above figure?

- 1 The third nucleotide on the anticodon may be modified to complementary base pair to different nucleotides.
- 2 A base-pair substitution at the third nucleotide of a triplet can result in the same amino acid being coded for.
- 3 Less than 20 different aminoacyl-tRNA synthetases are required to code for the naturally occurring amino acids.
- 4 All amino acids are coded for by more than one codon.
- 5 The genetic code is redundant but not ambiguous.

**A. 1, 2 and 5**

B. 1, 3 and 4

C. 2, 3 and 5

D. 1, 2, 4 and 5

### QUESTION 14

The following statements describe gene mutation.

- 1 It can occur in both somatic and sex cells.
- 2 It can cause sickle-cell anemia and Down syndrome in humans.
- 3 It can change the number of base pairs in a gene.
- 4 It can change a dominant allele into a recessive allele, but not a recessive allele to dominant allele.

Which statements are **not** correct?

A. 3 and 4

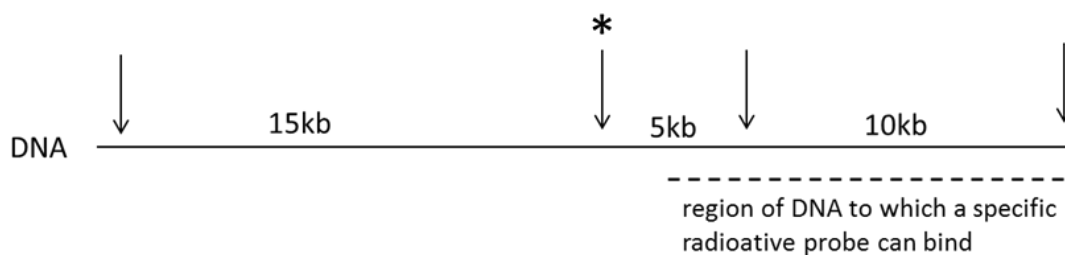
**B. 2 and 4**

C. 1 and 3

D. 1, 2 and 4

### QUESTION 15

A length of DNA from one of a pair of homologous chromosomes is shown. The target sites of *Eco*RI are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

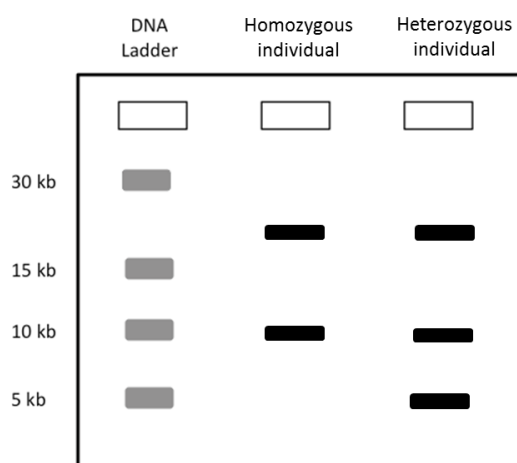


A mutation alters one base of the coding sequence of the site marked with an asterisk (\*). This also results in the loss of a target site for *Eco*RI.

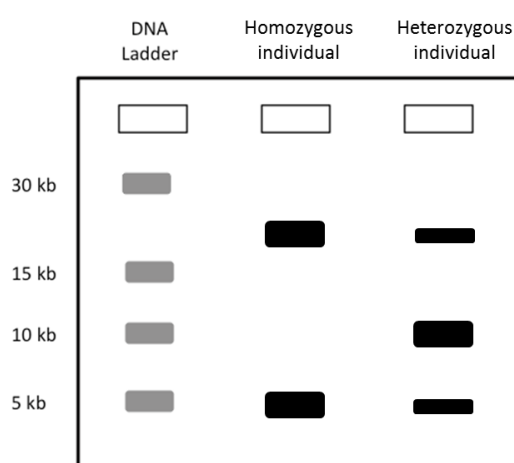
DNA from two individuals are cut with *Eco*RI and the DNA fragments separated according to size, and viewed subsequently by autoradiography.

Which of the following corresponds to the band patterns for individuals who are homozygous and heterozygous for this mutation respectively?

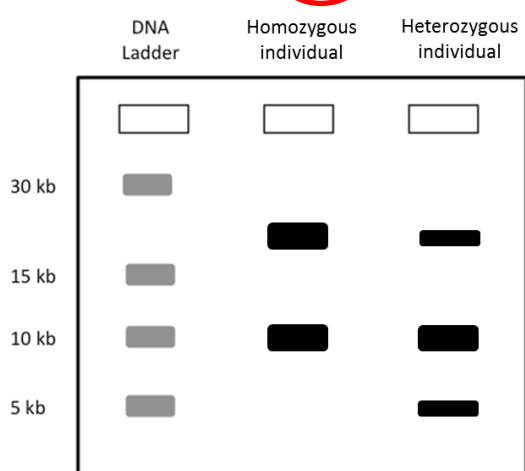
**A**



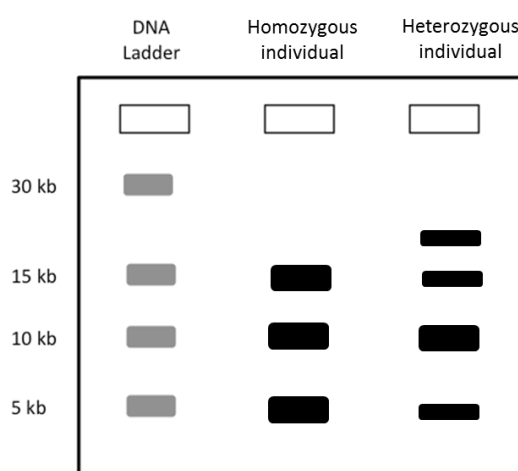
**B**



**C**

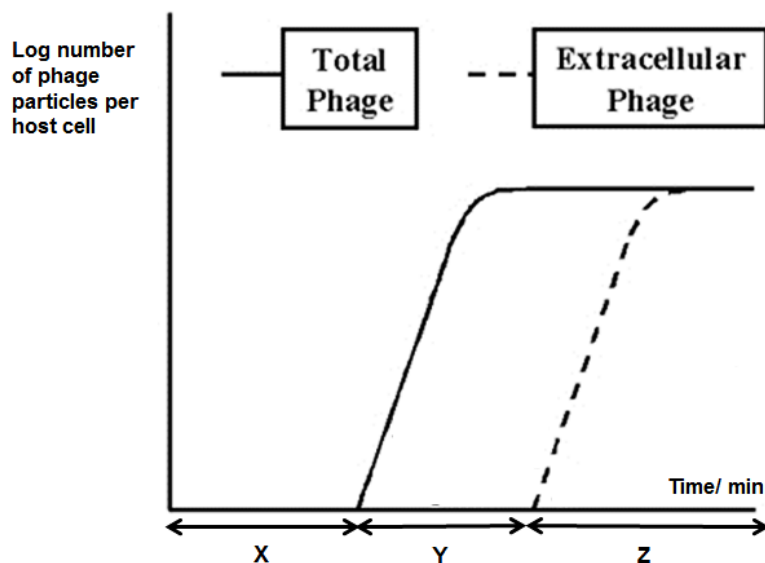


**D**



### QUESTION 16

The figure below shows a growth cycle of bacteriophages.



Which of the following is **true** about X, Y and Z of the growth cycle for T4 bacteriophage and lambda phage?

|    | T4 bacteriophage                                                 | Lambda phage                                                                                   |
|----|------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| A. | Period X is when the phage injects its viral RNA into host cell. | Period X is when the phage infects host cell and integrates its viral DNA into host chromosome |
| B. | Period Z is when phage lysozymes digest the host's cell wall.    | Cell lysis occurs in Period Z.                                                                 |
| C. | Period X is when hydrolysis of host cell occur.                  | Period X is where the prophage replicates.                                                     |
| D. | Period Y is when host cell's DNA is hydrolysed into fragments    | Period Y is when there is phage assembly.                                                      |

### QUESTION 17

Human immunodeficiency virus (HIV) is a retrovirus. After infecting a host cell, viral DNA is produced which is incorporated into the DNA of the host cell. The modified host genome now codes for the production of new HIV particles.

Which could be used as a potential treatment to slow down the spread of HIV?

- 1 Inhibitors of restriction endonucleases
- 2 Inhibitors of reverse transcriptase
- 3 Reverse transcriptase
- 4 (–) single-stranded RNA of HIV

A. 2 only

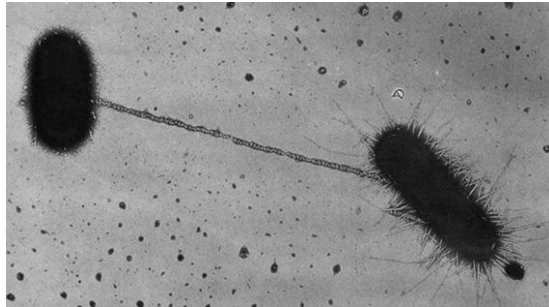
B. 1 and 2

C. 1 and 3

D. 2 and 4

### QUESTION 18

The photomicrographs below show two different processes occurring in bacteria.



Which of the following statements are **false**?

- 1 Both requires a protein appendage to take place.
- 2 In both process, semi-conservative replication of DNA occurs.
- 3 In both processes, replication of the bacterial chromosomal DNA occurs.
- 4 Both involved the transfer of a single-stranded DNA to another bacterial cell.

A. 1 and 2

B. 3 and 4

C. 1, 2 and 3

D. 1, 3 and 4

### QUESTION 19

Malvidin is a plant pigment responsible for the colours of red grapes, cranberries and blueberries. The dominant allele, **M**, codes for an enzyme involved in the biosynthesis of malvidin. The presence of dominant allele, **D**, of another unlinked gene, results in the absence of malvidin production in plants, even when the enzyme is present whilst the recessive allele, **d**, does not affect malvidin production.

A plant heterozygous at both loci was self-pollinated and gave rise to the following progeny:

|                                    |     |
|------------------------------------|-----|
| Plants with no malvidin production | 160 |
| Plants with malvidin production    | 40  |

The formula for the chi-squared ( $\chi^2$ ) test is given as follows:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

| degrees of freedom | probability |
|--------------------|-------------|
|                    | 0.05        |
| 1                  | 3.84        |
| 2                  | 5.99        |
| 3                  | 7.82        |
| 4                  | 9.49        |

Which conclusions may be drawn?

- 1 The expected phenotypic ratio for the self-pollination is 15:1.
- 2 The expected phenotypic ratio for the self-pollination is 3:1.
- 3 Difference between the observed and expected results is not significant.
- 4 The two genes controlling flower colour assort independently.
- 5 The difference is due to some factor such as linkage of the genes concerned.

A. 1, 4 and 5

B. 2, 3 and 4

C. 3 and 5

**D. 3 and 4**

### QUESTION 20

Which of the following statement(s) is/are true with regards to cyclic and non-cyclic photophosphorylation?

- 1 Only cyclic photophosphorylation produces oxygen.
- 2 Only cyclic photophosphorylation can function in the absence of photosystem II.
- 3 Only non-cyclic photophosphorylation will be affected in the absence of NADP reductase.
- 4 The plant switches from cyclic to non-cyclic photophosphorylation when only ATP is required.

A. 1 only

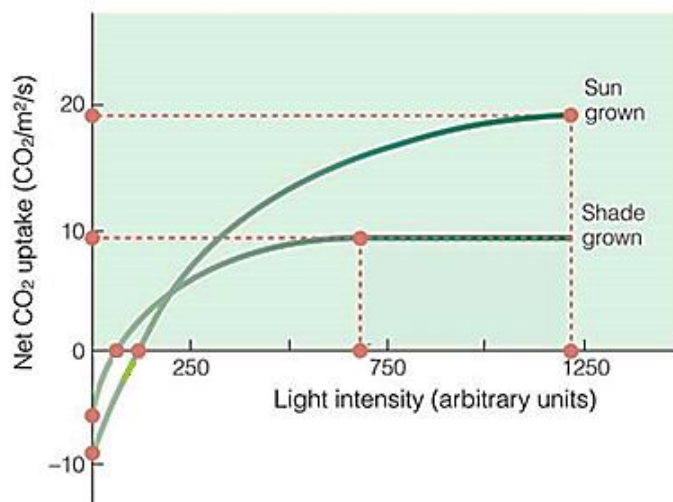
B. 1 and 4 only

**C. 2 and 3 only**

D. 2 and 4 only

### QUESTION 21

The effect of light intensity on photosynthetic rate was investigated in sun-grown and shade-grown leaves. The results obtained from this investigation are shown in the graph below.



Which of the following statement is a conclusion that can be drawn from the graph?

- A. There are more chloroplast-containing cells in sun-grown leaves than shade-grown leaves, thus light saturation point for sun-grown leaves is higher.
- B. Shade-grown leaves are more efficient at harnessing light energy at high light intensity.
- C. Compensation point of sun-grown leaves is higher than shade-grown leaves as sun-grown leaves require less carbon dioxide to carry out photosynthesis.
- D. Rate of Calvin cycle is faster in sun-grown leaves than shade-grown leaves at very low light.

### QUESTION 22

An experiment was conducted to investigate respiration of yeast cells.

**Tube 1:** Radioactive glucose solution + suspension of yeast cells + oxygen

**Tube 2:** Radioactive glucose solution + suspension of yeast cells + oxygen + antimycin

All the six carbon atoms of the radioactive glucose were <sup>14</sup>C. The initial radioactivity measured in each test tube was 60 arbitrary units.

Antimycin is an electron transport chain inhibitor.

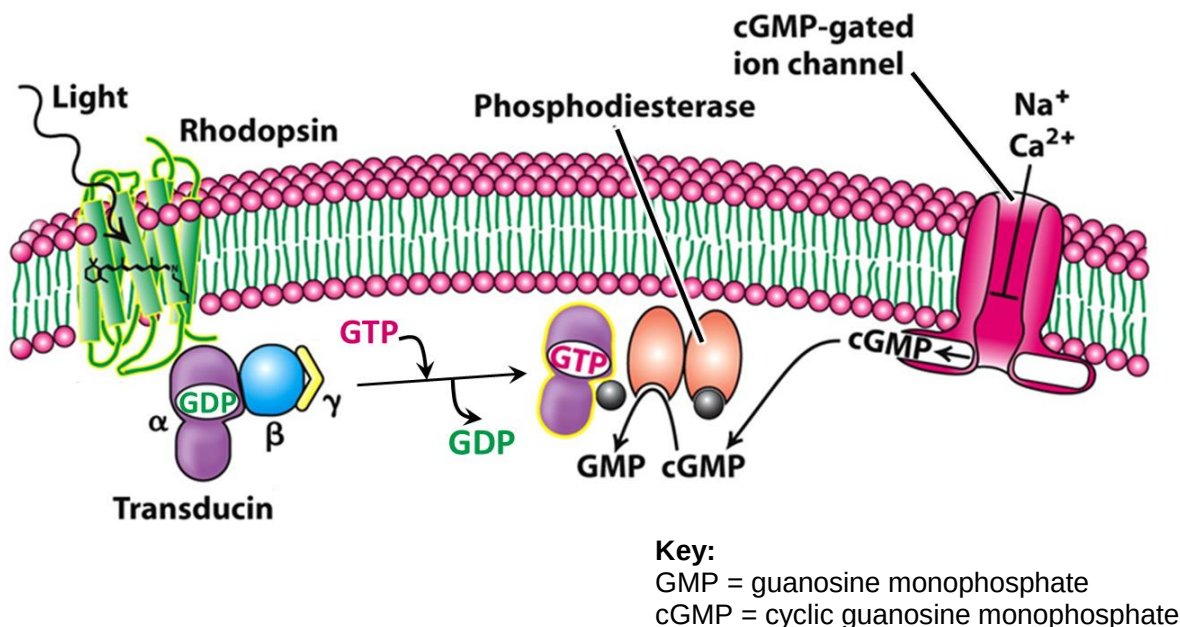
After all the glucose was metabolized, the amount of radioactivity in the gaseous product and the content of the tubes were measured. Which of the following shows the expected result?

|   | tube 1<br>(radioactivity / arbitrary units) |                 | tube 2<br>(radioactivity / arbitrary units) |                 |
|---|---------------------------------------------|-----------------|---------------------------------------------|-----------------|
|   | Content in tube 1                           | gaseous product | Content in tube 2                           | gaseous product |
| A | 40                                          | 20              | 0                                           | 60              |
| B | 0                                           | 60              | 20                                          | 40              |
| C | 0                                           | 60              | 40                                          | 20              |
| D | 60                                          | 0               | 40                                          | 20              |

### QUESTION 23

Vision is based on the absorption of light by photoreceptor cells in the eye. Detection of light by the photoreceptor cells is mediated by a transmembrane receptor protein, rhodopsin. Absorption of light by rhodopsin initiates a cascade of events that closes an ion-channel, resulting in a change the voltage (difference in charges) across the cell membrane, thus producing a signal which is communicated to the brain.

The figure below illustrates the signaling events that take place in a photoreceptor cell upon light stimulation.



Which of the following correctly describes the role of the proteins in rhodopsin signaling?

|           | Rhodopsin                                                                        | Transducin                                                                     | Phosphodiesterase                                                                        | cGMP-gated ion channel                                                                     |
|-----------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>A.</b> | a G-protein linked receptor which changes conformation upon light absorption     | a G-protein which is activated when the bound GDP replaced by GTP              | activated by GTP-bound transducin and converts cGMP to GMP to terminate the transduction | closes when cGMP dissociates from it, preventing ions from entering the photoreceptor cell |
| <b>B.</b> | a G-protein linked receptor which changes conformation upon light absorption     | a relay protein which is activated when the bound GDP replaced by GTP          | converts cGMP to CMP, which is a second messenger that brings about a response           | closes when cGMP dissociates from it, preventing ions from entering the photoreceptor      |
| <b>C.</b> | a G-protein linked receptor which changes conformation upon binding to G protein | a G-protein which is activated when the bound GDP is phosphorylated to GTP     | activated by GTP-bound transducin and converts cGMP to GMP to terminate the transduction | opens when cGMP dissociates from it, allowing ions to enter the photoreceptor cell         |
| <b>D.</b> | a G-protein linked receptor which changes conformation upon binding to G protein | a relay protein which is activated when the bound GDP is phosphorylated to GTP | converts cGMP to CMP, which is a second messenger that brings about a response           | opens when cGMP dissociates from it, allowing ions to enter the photoreceptor cell         |



**QUESTION 24**

Which statements are proposed by the Darwinian evolutionary theory?

1. Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
2. In competition for survival, the more aggressive animals are more likely to survive.
3. An individual most adapted to a stable environment will stop evolving.
4. Variation between individuals of a species is essential for evolutionary change.

**A.** 1, 2 and 4 only    **B.** 2 and 3 only    **C.** 3 and 4 only    **D. 4 only**

**QUESTION 25**

Human activity often results in habitat loss. The remaining habitat in an area become fragmented forming smaller patches of habitat, through for example, construction of new roads and deforestation.

Which statements describe how a small habitat patch differs from a larger patch of the same habitat?

1. biodiversity decreases
2. competition from surrounding habitats increases
3. gene pool increases
4. populations of large animals decrease

**A.** 1 and 2 only  
**B.** 2 and 3 only  
**C.** 3 and 4 only  
**D. 1, 2 and 4 only**

### QUESTION 26

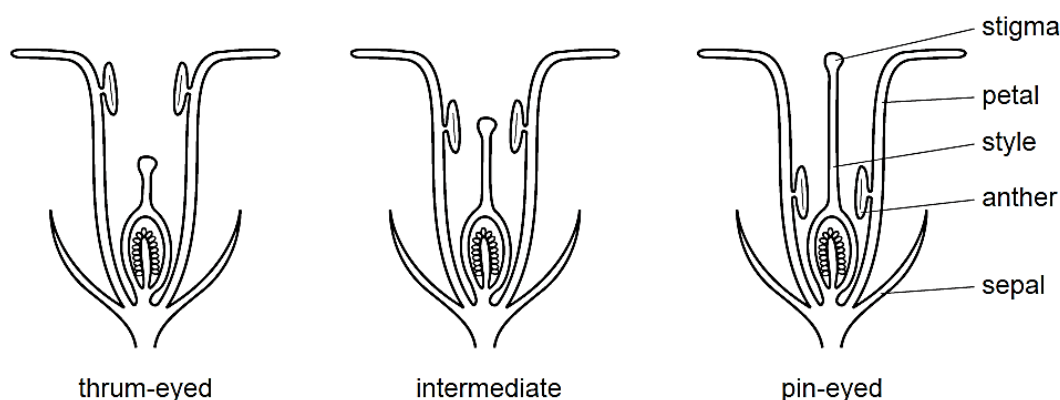
The primrose, *Primula vulgaris*, is a small herbaceous, yellow-flowered plant which is common in cooler areas of the Northern hemisphere including alpine and Arctic areas.

The flowers of the primrose have different flower shapes (polymorphic), which are adaptations for pollination. 'Thrum-eyed' primroses have a short style. 'Pin-eyed' primroses have much longer styles. The anther position also varies among the primrose.

Some populations of primrose consist almost entirely of plants with intermediate flowers. These populations are common where there are fewer winged insects.

Anthers produce pollen (male gametes) which land on the stigma, leading to fertilization.

The diagrams show polymorphic flowers of primroses.



Which statements are correct?

1. Cross-pollination will be favoured between pin-eyed and thrum-eyed primroses.
2. Primroses with pin-eyed flowers are likely to show more genetic diversity than primroses with intermediate flowers.
3. Primroses with thrum-eyed flowers are likely to be more able to adapt to changing environmental conditions than pin-eyed primroses.
4. Self-pollination is more likely to occur in primroses with intermediate flowers.

A. 1 and 2 only      B. 1, 2, 3 and 4      **C. 1, 2 and 4 only**      D. 3 and 4 only

### QUESTION 27

Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome c. Cytochrome c is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the ideas that

- all living organisms are related, and
  - there is a single, rather than a multiple, origin of life?
1. The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
  2. The sequence of amino acids in cytochrome c is similar in organisms that are from similar environments or with similar metabolic demands.
  3. The majority of organisms have the same, or similar, amino acid sequences for cytochrome c.
  4. When transferred into a very dissimilar organism, a gene coding for cytochrome c will lead to the expression of a protein that will function in the other organism.

A. 1 and 2 only      B. 2 and 3 only      **C. 3 and 4 only**      D. 1, 3 and 4 only

### QUESTION 28

Which statement about immunity is correct?

- A. Antibody donation, but not antibody production, occurs with artificial active and artificial passive immunity.
- B. Artificial active immunity lasts for a greater length of time than natural passive immunity.**
- C. Natural active immunity provides a faster response to infection than artificial active immunity.
- D. Recognition and binding by specific B-lymphocytes only occurs with natural immunity.

### QUESTION 29

A student wrote down five statements about antibodies.

1. Their structure depends on peptide, hydrogen and disulfide bonds.
2. They are protein molecules with both tertiary and quaternary structure.
3. Four polypeptides are coded for by two different genes.
4. The great variation in antigen specificity is a result of alternative RNA splicing.
5. Four polypeptides provide four antigen binding sites of the same specificity.

Which statements are true?

**A. 1, 2 and 3 only**      B. 1, 3 and 4 only      C. 2, 4 and 5 only      D. 2, 3 and 5 only

**QUESTION 30**

Forests usually provide habitats for a great number of species. The loss of species from ecosystems as a result of anthropogenic climate change is likely to affect food webs. However, ascertaining how the removal of one species from a food web might affect others is a challenge.

Which of the following statements explain why it might be difficult to ascertain such effects?

1. The loss of one species might affect multiple connections in food web.
2. Organisms can switch their diet when their primary food source is scarce.
3. The consequences on a food web might take a long time to occur.
4. It is difficult to identify trophic levels in a food chain because of the diverse feeding behaviours.

**A. 1, 2, 3 and 4**

**B. 1, 2 and 3 only**

**C. 2 and 4 only**

**D. 3 and 4 only**

• THE END •